

- 5) Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
shanmukhi@ShanmukhiPC: ~  
$ nano producer_consumer.c  
$ gcc producer_consumer.c -o producer_consumer  
$ ./producer_consumer  
Child received: Hostel Room Data: Room 101 - Occupied  
  
Communication completed.  
Communication time: 482.62 microseconds  
$ nano pipeline.c  
$ gcc pipeline.c -o pipeline  
$ ./pipeline  
drwxr-xr-x 8 shanmukhi shanmukhi 4096 Aug 3 05:02 OS_Project  
drwxr-xr-x 2 shanmukhi shanmukhi 4096 Aug 4 09:31 Practical1  
drwxr-xr-x 2 shanmukhi shanmukhi 4096 Aug 4 10:02 Practical2  
drwxr-xr-x 2 shanmukhi shanmukhi 4096 Aug 5 06:48 Practical3  
-rw-r--r-- 1 shanmukhi shanmukhi 1102 Sep 16 06:35 pipeline.c  
-rwxr-xr-x 1 shanmukhi shanmukhi 16400 Sep 16 06:34 producer_consumer  
-rw-r--r-- 1 shanmukhi shanmukhi 1155 Sep 16 06:34 producer_consumer.c  
$ ls  
Lab OSLab OS_Project Practical2 lab pipeline producer_consumer  
OS OS_Lab Practical1 Practical3 os pipeline.c producer_consumer.c  
$ ./pipeline  
drwxr-xr-x 8 shanmukhi shanmukhi 4096 Aug 3 05:02 OS_Project  
drwxr-xr-x 2 shanmukhi shanmukhi 4096 Aug 4 09:31 Practical1  
drwxr-xr-x 2 shanmukhi shanmukhi 4096 Aug 4 10:02 Practical2  
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$ |
```